+1 540-558-3749
Blacksburg, Virginia
ragul.seeth@gmail.com

RAGUL SEETHARAMAN

LinkedIn:
https://www.linkedin.com/in/ragul-seetharaman

Education

Virginia Tech, USA *[Dual Masters] MSc in Computer Engineering & MSc in Biological Systems Engineering* **Aug 2023 – May 2026**
Vellore Institute of Technology, India, *BTech in Electronics and Communication Engineering* **Jul 2016 - May 2020**

Skills

Programming	Python, SQL, C++, Matlab
Tools/Frameworks	TensorFlow, PyTorch, OpenCV, LangChain, LangGraph, Airflow, FastAPI, AWS(SageMaker,EC2,S3), Fine-Tuning(LoRA/QLoRA), MCP, Retrieval-Augmented-Generation, Docker, Prompt Engineering
Coursework	Computer Vision | Advanced Machine Learning | Natural Language Processing | Database Systems & Management | Data Engineering | Applications of Machine Learning

Experience

Cotiviti, Utah (USA) | **Healthcare Payment Accuracy** **Jun 2025 — Present**
Generative AI Research Engineer Intern

- Developed a contextual RAG architecture using **PaddleOCR-VL** for page-level text extraction, semantic chunk segmentation, and embedding generation with the **OpenAI text-embedding-3-large** model, combining **Amazon S3 Vectors**-based dense retrieval with **BM25 via adaptive rank fusion** to power contract intelligence across 2,630 retail agreements and projected to save up to $4M annually.
- Built a **LangGraph**-based retrieval workflow with **stateful routing** to support **entity resolution, citation tracing**, and **multi-hop answer generation** across Master Service Agreements, Statements of Work, Amendments, Work Orders, and clause-level references.
- Deployed the service on AWS EC2 with Docker Compose, fronted by a **Load Balancer** and **Route 53** routing to enable secure, scalable access across the organization.

Turmerik, New York (USA) | **Biotechnology** **Sep 2024 — Dec 2024**
Machine Learning Engineer Intern | *NSF I-CORPS*

- Developed a clinical trial protocol generation system using **LangChain**, indexing embeddings in **ChromaDB**, retrieving similar studies with **FAISS**, and using **ROUGE**-scoring evaluation to improve output quality from ClinicalTrials.gov data.
- Led customer discovery as Entrepreneurial Lead in the **NSF I-CORPS** program, engaging 25+ industry leaders to align product development with clinical research needs.

AmberTAG Analytics (nference), USA (remote) | **Healthcare** **May 2021 — Nov 2022**
Data Analyst

- Improved heart arrhythmia detection accuracy to 94.2% by optimizing **1D CNN** and **Xception** models on ECG/EEG data with **Fourier Transform** and **Empirical Mode Decomposition (EMD)** for noise reduction, while supporting a **RabbitMQ**, **PostgreSQL**, and 3D visualization pipeline tested on 11 patients, contributing to the product's **FDA** approval and acquisition by nference.

Cognizant, India | **E-commerce** **Sep 2020 — May 2021**
Programmer Analyst Trainee

- Developed a Shopify recommendation and analytics system by vectorizing product descriptions with **TF-IDF** and **Word2Vec**, clustering products using **K-modes** for personalized recommendations, and automating sales reporting for 6,000+ products with **AWS SageMaker**

Patents & Research

ADVANCED SHOPPING TROLLEY WITH PRODUCT AUTHENTICATION AND FAST BILLING **Sep 2020**
Patent no: 202041036031, 20204102288 | *IEEE CONECCT International Conference* | *National Finalist (Top 1.7%) by Larsen & Toubro*

- Developed an IoT self-checkout system using Raspberry Pi, integrating **YOLOv2 Tiny** and **OpenCV** for 95.3% accurate **object detection** and barcode scanning with load cells for automated billing
- Built a **Firebase**-powered admin interface for real-time monitoring of 40+ trolleys (scalable to SQL) and implemented collaborative filtering to deliver personalized recommendations, reducing customer queue time by 8 minutes.

Projects

Crediwise – Use the right card, every time. **Sep 2025**
MLH Hackathon – Capital One: Best Financial Hack | *VT StartupSprint: Best Problem-Solution Fit*

- Developed a full-stack system with a **React** frontend, **Flask** backend, and **Manifest V3** Chrome extension, integrated with **Supabase** to deliver real-time checkout **card recommendations** and an interactive savings dashboard.
- Integrated Bright Data **MCP agent** to scrape live credit card offers and began adding RAG retrieval to reduce hallucinations in search results.

Agentic Self-Corrective Retrieval-Augmented Generation **Oct 2024**
Advanced AI Pro Bootcamp

- Designed an adaptive RAG pipeline with **LangChain** and **ChromaDB** for document and URL retrieval, integrating Google Custom Search API to reduce hallucinations and improve answer robustness.
- Deployed on AWS EC2 with **Docker Compose** and a Streamlit UI, powered by a 4-bit quantized **LLaMA-3** model fine-tuned on an **Ollama** server, improving system relevance by 20%